

## Assignment 1 Submission; due Thursday, September 24, 2026

Name: Vlad Paun

Student number: 21291453

- [1] Let  $E \subseteq \mathbb{R}$  be nonempty. We say that  $t \in \mathbb{R}$  is a *lower bound* of  $E$  if  $t \leq x$  for all  $x \in E$ . We say that  $E$  is *bounded below* if there exists a lower bound of  $E$ . We say that  $t \in \mathbb{R}$  is a *greatest lower bound*, or *infimum*, of  $E$  if it is a lower bound of  $E$  and if  $t \geq s$  for any lower bound  $s$  of  $E$ . If  $E$  has an infimum, it is denoted  $\inf E$ .

- [a] Let  $E$  be a nonempty subset of  $\mathbb{R}$  which is bounded below. Find a way to use the completeness axiom to prove that  $\inf E$  exists. *Hint:* Consider the subset  $-E = \{-x : x \in E\}$ .

**Answer:**

*Proof.* Since  $E$  is nonempty and bounded below, there exists  $\alpha \in \mathbb{R}$  such that

$$\forall x \in E : \alpha \leq x$$

Consequently,

$$\forall x \in E : -\alpha \geq -x$$

.

Define

$$-E = \{-x : x \in E\}.$$

Then,  $-\alpha$  is an upper bound of  $-E$ .

By the completeness axiom,  $\sup(-E)$  exists. Let

$$\beta = \sup(-E).$$

It follows that  $\forall x \in -E : \beta \geq x$ . And therefore,  $\forall x \in E : -\beta \leq x$ . So  $-\beta$  is a lower bound of  $E$ .

Next, take any other lower bound of  $E$ , call it  $s$ . From the same logic as above, it follows that  $-s$  is an upper bound of  $-E$ . Since  $\beta$  is a supremum of  $-E$ , it follows that  $\beta \leq -s$ .

So,  $-\beta \geq s$ . Therefore, since  $-\beta$  is a lower bound of  $E$  and it is at least as large as any other lower bound, it follows that  $-\beta = \inf E$   $\square$

- [b] Let  $\emptyset \neq A \subseteq B \subseteq \mathbb{R}$ . Suppose  $B$  is bounded below (so  $\inf B$  exists). Prove that  $A$  is bounded below (so  $\inf A$  exists) and that  $\inf A \geq \inf B$ .

**Answer:**

*Proof.* If  $B$  is bounded below, then  $\exists \alpha : \forall x \in B : \alpha \leq x$ .

Since  $A \subseteq B$ , we know that  $\forall y \in A : y \in B$ .

Consequently,

$$\forall y \in A : \alpha \leq y$$

Therefore,  $\alpha$  is also a lower bound of  $A$ . And from 1[a], we can conclude that  $\inf A$  also exists.

Now, since  $A$  is a subset of  $B$  and  $\inf B$  is a lower bound of  $B$ , then  $\inf B$  must also be a lower bound of  $A$ . Since  $\inf A$  must be greater than or equal to any other lower bounds of  $A$ , we have that

$$\inf A \geq \inf B$$

□

- [c] Let  $A, B$  be nonempty subsets of  $\mathbb{R}$ , both bounded below (so both  $\inf A$  and  $\inf B$  exist). Define

$$A + B = \{a + b : a \in A, b \in B\}.$$

Prove that  $A + B$  is bounded below and that  $\inf(A + B) = \inf A + \inf B$ .

**Answer:**

*Proof.*

$$\begin{aligned} \forall a \in A : \inf A &\leq a \\ \forall b \in B : \inf B &\leq b \\ \therefore \forall a \in A, b \in B : \inf A + \inf B &\leq a + b \\ \therefore \forall x \in A + B : \inf A + \inf B &\leq x \end{aligned}$$

So  $\inf A + \inf B$  is a lower bound of  $A + B$ . Consequently,  $\inf(A + B)$  exists.

Now, we know that

$$\begin{aligned} \forall \varepsilon > 0 : \exists \alpha \in A : \inf A + \frac{\varepsilon}{2} &> \alpha \\ \forall \varepsilon > 0 : \exists \beta \in B : \inf B + \frac{\varepsilon}{2} &> \beta \end{aligned}$$

Therefore,

$$\forall \varepsilon > 0 : \exists \alpha \in A, \beta \in B : \inf A + \inf B + \varepsilon > \alpha + \beta$$

This means that no number greater than  $\inf A + \inf B$  can be a lower bound of  $A + B$ . Therefore,

$$\inf(A + B) = \inf A + \inf B$$

□

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- [2] Let  $x, y$  be two real numbers such that  $y - x > 1$ . (In particular,  $x < y$ .)

- [a] If  $x$  is positive, prove that there exists an integer  $k \in \mathbb{Z}$  such that  $x < k < y$ .

**Answer:**

*Proof.* By the Archimedean property, the set  $E = \{n \in \mathbb{N} : n > x\} \subset \mathbb{N} \subset \mathbb{Z}$  is not empty.

$x$  is a lower bound of  $E$ , so by the completeness axiom,  $\inf E$  exists. And since  $E \subset \mathbb{Z}$ ,  $\inf E \in E$ . Let  $\inf E = k$ .

Then  $x < k$ . Now, assume for the sake of contradiction that  $x < k - 1$ . Since  $k - 1 \in \mathbb{Z}$ , then  $k - 1 \in E$ . But  $k - 1 < k$  and  $k = \inf E$ , which is a contradiction.

Therefore,  $k - 1 \leq x$ .  $k \leq x + 1 < y$ .

So,  $x < k < y$ . □

- [b] If  $x$  is negative and  $y$  is positive, prove that there exists an integer  $k \in \mathbb{Z}$  such that  $x < k < y$ .

**Answer:**

*Proof.* Take  $k = 0 \in \mathbb{Z}$ . □

- [c] If  $y$  is negative, prove that there exists an integer  $k \in \mathbb{Z}$  such that  $x < k < y$ .

**Answer:**

*Proof.* Since  $y < 0$ , we have  $-y > 0$ . Moreover,

$$(-x) - (-y) = y - x > 1$$

So applying 2[a] to  $-x$  &  $-y$ , we get that  $\exists n \in \mathbb{Z} : -y < n < -x$ .

$$x < -n < y$$

Taking  $k = -n \in \mathbb{Z}$  proves the result. □

- [3] The parts of this problem are *not related*.

- [a] Let  $x \in \mathbb{R}$ . Consider the set  $E = \{q \in \mathbb{Q} : q < x\} \subset \mathbb{Q} \subset \mathbb{R}$ . Show that  $E$  is nonempty and bounded above, so  $\sup E$  exists. Show also that  $\sup E = x$ .

**Answer:**

*Proof.* Applying the Archimedean property to  $-x$  we get that  $\exists n \in \mathbb{N} : n > -x$ . So,  $x > -n$ . Therefore,  $-n \in E$ , so  $E$  is non-empty.

Also, since  $x$  is strictly larger than any element of  $E$ ,  $x$  is a lower bound of  $E$ .

Since  $E$  is bounded above and non-empty,  $\sup E$  exists.

Let  $\sup E = \alpha$ . We know that  $\alpha \leq x$ . Now, assume for the sake of contradiction, that  $\alpha < x$ .

One of the propositions proved in class showed that between any two distinct real numbers, there exists a rational number. That is:

$$\exists q \in \mathbb{Q} : \alpha < q < x$$

But if  $q$  is rational and smaller than  $x$ , then it should be a part of  $E$ . But this contradicts  $\alpha$  being an infimum of  $E$ . Therefore, our assumption must be false and

$$\sup E = x$$

□

- [b] Let  $x, y \in \mathbb{R}$  with  $x < y$ . Prove that there exists an *irrational* (not rational) number  $z$  such that  $x < z < y$ . *Hint:* You may use the fact that  $\sqrt{2}$  is irrational.

**Answer:**

*Proof.* We will take the interval  $(x, y)$  and shift it by  $\sqrt{2}$ .

$$x - \sqrt{2} < y - \sqrt{2}$$

By the proposition shown in class, we know that

$$\exists q \in \mathbb{Q} : x - \sqrt{2} < q < y - \sqrt{2}$$

Therefore,  $x < q + \sqrt{2} < y$ . Since we know that  $\sqrt{2}$  is irrational, then  $z = q + \sqrt{2}$  is also irrational. □

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- [4] Let  $C > 0$  be a fixed positive number.

- [a] Let  $x \geq 0$ , and suppose that for any  $\varepsilon > 0$ , we have  $x < C\varepsilon$ . Show that for any  $\eta > 0$ , we have  $x < \eta$ .

**Answer:**

- [b] In the situation of part [a], show that we must have  $x = 0$ . *Hint:* Find a way to use the Archimedean property.

**Answer:**

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- [5] COMING SOON

**Answer:**

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- [6] COMING SOON

**Answer:**

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[7] COMING SOON

**Answer:**

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[8] COMING SOON

**Answer:**

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[9] COMING SOON

**Answer:**

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[10] COMING SOON

**Answer:**